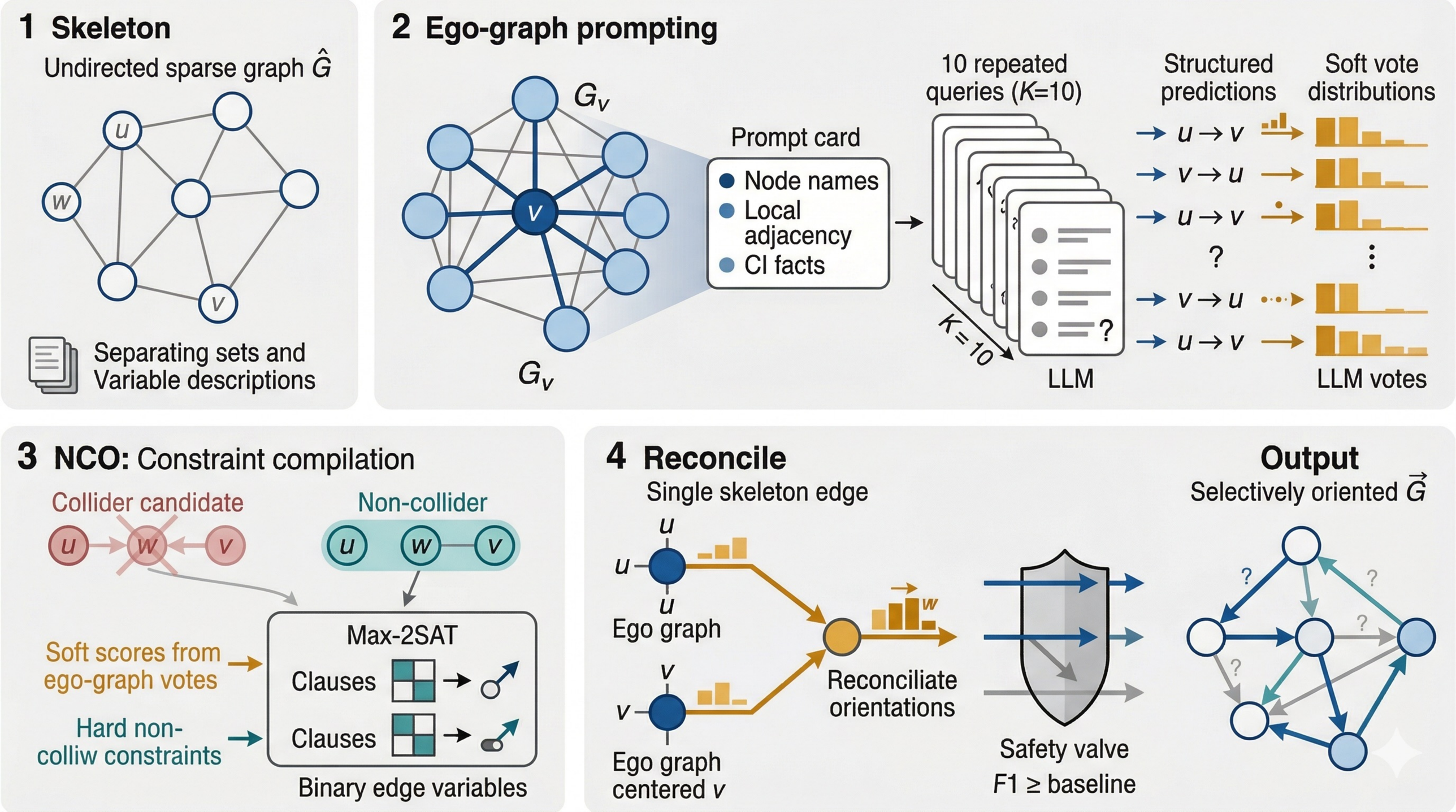
1 Skeleton
Undirected sparse graph $\hat{G}$
u
w
v
Separating sets and
Variable descriptions
2 Ego-graph prompting
G_v
v
G_v
Prompt card
Node names
Local adjacency
CI facts
10 repeated queries (K=10)
K=10
?
LLM
Structured predictions
$u \to v$
$v \to u$
$u \to v$
?
$v \to u$
$u \to v$
Soft vote distributions
LLM votes
3 NCO: Constraint compilation
Collider candidate
u
w
v
Non-collider
u
w
v
Max-2SAT
Soft scores from ego-graph votes
Clauses
Hard non-colliw constraints
Clauses
Binary edge variables
4 Reconcile
Single skeleton edge
u
u
u
Ego graph
v
v
Ego graph centered v
w
Reconciliate orientations
Safety valve
$F1 \geq$ baseline
Output
Selectively oriented $\vec{G}$
?
?
?
?